EXPERIMENT No.2

REALIZATION OF HALF ADDER AND FULL ADDER

AIM: To realize Half Adder and Full Adder using basic gates.

APPARATUS:

Breadboard, IC 7486(XOR), IC 7408 (AND), IC 7432(OR), LEDs, 5V power supply, connecting wires.

THEORY:

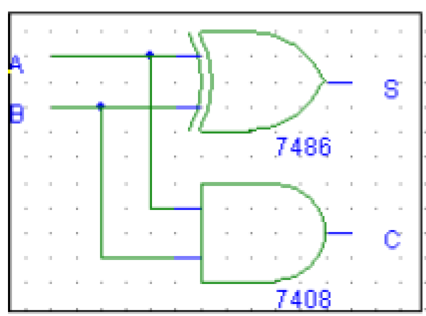
HALF ADDER- A combinational logic circuit that performs the addition of two data bits, A and B, is called a half-adder. Addition will result in two output bits; one of which is the sum bit, S, and the other is the carry bit, C. The Boolean functions describing the half-adder are:

S =A □□B

$C = A\,B$

TRUTH TABLE

INPUTS		OUTPUTS	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



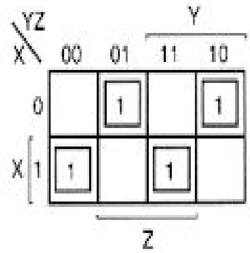
FULL ADDER- The half-adder does not take the carry bit from its previous stage into account. This carry bit from its previous stage is called carry-in bit. A combinational logic circuit that adds two data bits, A and B, and a carry-in bit, Cin , is called a full-adder

$$S = (x \oplus y) \oplus Cin$$

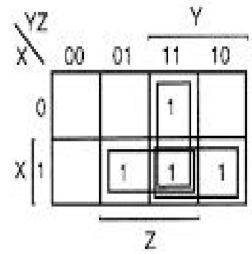
$$C = xy + Cin\,(x \oplus y)$$

TRUTH TABLE

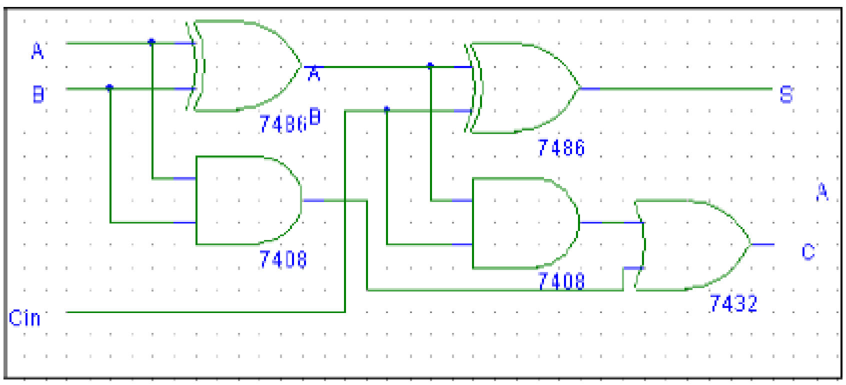
INPUTS			OUTPUTS	
A	B	Cin	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



$$S = \bar{X}\bar{Y}Z + \bar{X}Y\bar{Z} + X\bar{Y}\bar{Z} + XYZ$$
$$= X \oplus Y \oplus Z$$



$$C = XY + XZ + YZ$$
$$= XY + Z(X\bar{Y} + \bar{X}Y)$$
$$= XY + Z(X \oplus Y)$$



PROCEDURE:

1. Verify the gates.

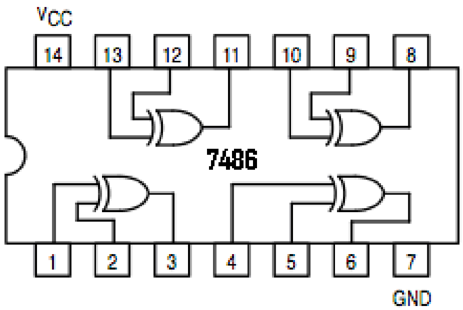
2. Make the connections as per the circuit diagram.

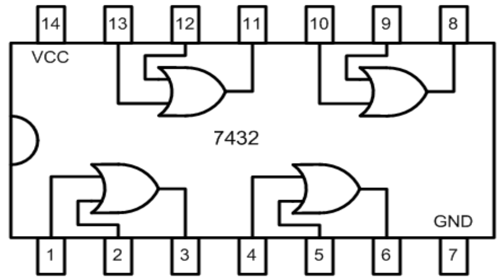
3. Switch on Vcc and apply various combinations of input according to truth table.

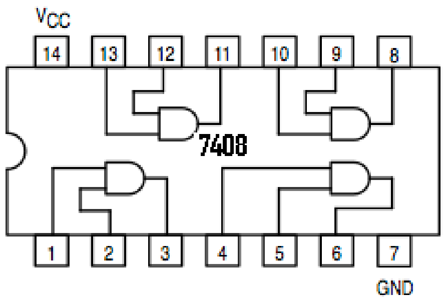
4. Note down the output readings for half adder and full adder.

Sum and the carry bit for different combinations of inputs verify their truth tables.

Pin Configuration of IC-7486(XOR), IC-7432(OR), IC-7408(AND):







PRECAUTIONS:

1. All ICs should be checked before starting the experiment.
2. All the connection should be tight.
3. Always connect ground first and then the supply.
4. Switch off the power supply after completion of the experiment.

RESULT:

Half Adder and Full Adder have been realized using basic gates and their truth table has been verified.

VIVA-VOCE QUESTIONS:

1. Implement half adder and Full adder with NAND Gate only.
2. Implement half adder and Full adder with NOR Gate only.
3. Prove that Sum of a full adder is a XOR Gate between its inputs.
4. State the difference between Half Adder and Full Adder.
5. Design a full adder using two half adders.